

ESTIMATION OF PARAMETERS OF A TRUNCATED MULTIVARIATE NORMAL DISTRIBUTION

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SUMMARY

In this paper, we consider an extension of results of Dyer (1973) and Nath (1971) on the estimation of parameters of a doubly truncated bivariate normal distribution to the estimation of parameters of a doubly truncated multivariate normal distribution. Recurrence relations for the moments of a doubly truncated p-variate normal distribution are derived. Methods of computation of the estimates of parameters based on a sample moments and on maximum likelihood are developed. It is shown that the results of Dyer and Nath regarding the estimates of doubly truncated bivariate normal distribution can be deduced as special cases.

1. Introduction

Khatri and Jaiswal (1963) have given the moment and maximum likelihood estimators of a bivariate normal distribution truncated from the lower end with respect to both variables. Nath (1971) has obtained the maximum likelihood equations of a doubly truncated bivariate normal distribution and the likelihood estimates of the parameters can be obtained by solving the equations by the method of iteration. Recently Dyer (1973) has obtained the moment estimators for the parameters of the doubly truncated bivariate normal distribution. This paper deals with the estimation of parameters of a p-variate normal distribution, doubly truncated with respect to the p variables. In quality control analysis, a p-variate doubly truncated normal distribution arises when a lot of the products with multiple characters is inspected with respect to both lower and upper specification limits. It is also shown that the method suggested by Dyer (1973) to obtain the system of linearly independent equations (particularly the third equation in (7) of Section 3) works well in the bivariate case only but it cannot be extended in the case of p-variate normal distributions. The likelihood equations in the p-variate case are derived and those obtained by Nath (1971) can be deduced as particular cases. A numerical example is given to illustrate some of the results obtained in the paper.

2. Recurrence Relations for Population Moments

The density function of a p -dimensional random variable $x' = (x_1, \dots, x_p)$ having p -variate normal distribution, doubly truncated with respect to all variables, is given by

$$(1) \quad f(x) = g(x)/G \quad a_{1i} \leq x_i \leq a_{2i}, i = 1, 2, \dots, p,$$

where

$$(2) \quad g(x) = (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp\left(-\frac{1}{2} (x-\mu)' \Sigma^{-1} (x-\mu)\right)$$

and

$$(3) \quad G = \int_{a_{11}}^{a_{21}} \dots \int_{a_{1i}}^{a_{2i}} \dots \int_{a_{1p}}^{a_{2p}} g(x) dx_1 \dots dx_p.$$

The unknown parameters are μ and Σ and it is assumed that the truncation points (a_{1i}, a_{2i}) for $i = 1, 2, \dots, p$ are known.

Letting $y = x - a_1$ where $a_1' = (a_{11}, \dots, a_{1p})$, the density function of y is given by

$$(4) \quad f(y) = g(y + a_1)/G, \quad 0 \leq y_i \leq a_{2i} - a_{1i} = R_i, i = 1, 2, \dots, p.$$

Taking the first partial derivative with respect to y_i , we get

$$(5) \quad \frac{\partial f(y)}{\partial y_i} = \left\{ \sum_{j=1}^p h_{ij} y_j + \eta_i \right\} f(y), i = 1, 2, \dots, p$$

where

$$(6) \quad H = (h_{ij}) = -\Sigma^{-1}, i = 1, 2, \dots, p \text{ and } \eta = -\Sigma^{-1}(a_1 - \mu).$$

Multiplying both sides of (5), for $i = 1$ by $\left\{ \prod_{i=1}^p y_i^{r_i} \right\}$ and integrating over

$\{(y_1, \dots, y_p); 0 < y_i < R_i\}$, $i = 1, 2, \dots, p$ we obtain the relation for the population moments $\mu_{r_1, r_2, \dots, r_p}' = E \left\{ \prod_{i=1}^p y_i^{r_i} \right\}$ as follows:

$$\begin{aligned} & \int_0^{R_1} \dots \int_0^{R_p} \frac{\partial f(y_1, \dots, y_p)}{\partial y_1} y_1^{r_1} y_2^{r_2} \dots y_p^{r_p} dy_1 \dots dy_p \\ &= \int_0^{R_1} \dots \int_0^{R_p} \left(\sum_{j=1}^p h_{1j} y_j + \eta_1 \right) y_1^{r_1} \dots y_p^{r_p} \cdot f(y) dy_1 \dots dy_p \end{aligned}$$

Integrating the left hand side of the above equation partially with respect to y_1 , we get

$$\begin{aligned}
& R_1^{r_1} \int_0^{R_2} \dots \int_0^{R_p} f(R_1, y_1, \dots, y_p) y_2^{r_2} \dots y_p^{r_p} dy_2 \dots dy_p - r_1 \mu'_{r_1-1, r_2, \dots, r_p} \\
& = \sum_{j=1}^p h_{1j} \mu'_{r_1, \dots, r_{j-1}, r_j+1, r_{j+1}, \dots, r_p} + \eta_{r_1, \dots, r_p}
\end{aligned}$$

Rearrangement of terms yields

$$\begin{aligned}
(7) \quad r_1 \mu'_{r_1-1, r_2, \dots, r_p} &= R_1^{r_1} K(r_2, \dots, r_p) + \sum_{j=1}^p h_{1j} \mu'_{r_1, \dots, r_{j-1}, r_j+1, r_{j+1}, \dots, r_p} \\
&+ \eta_{r_1, r_2, \dots, r_p}
\end{aligned}$$

where

$$\begin{aligned}
(8) \quad K(r_2, \dots, r_p) &= \int_0^{R_2} \dots \int_0^{R_p} \frac{g(R_1 + a_{11}, y_2 + a_{12}, \dots, y_p + a_{1p})}{G} \cdot y_2^{r_2} \dots y_p^{r_p} \\
&\quad \cdot dy_2 \dots dy_p \\
&= \frac{\phi\left(\frac{a_{21} - \mu_1}{\sqrt{\sigma_{11}}}\right) |\Sigma_{22.1}|^{1/2}}{(2\pi)^{(p-1)/2} |\Sigma|^{1/2} G} \int_{\alpha_i < Z_i < \beta_i} \prod_{i=2}^p \left(\phi_i + \sum_{j=2}^p \ell_{ij} Z_j\right)^{r_i} \\
&\quad \cdot \exp\left(-\frac{1}{2} \sum_{i=2}^p Z_i^2\right) dz_2 \dots dz_p
\end{aligned}$$

$$y'_2 = (y_2, \dots, y_p), Z' = (Z_2, \dots, Z_p) = [\Sigma_{22.1}^{1/2} (y_2 - \phi_2)]', \Sigma_{22.1}^{1/2} = (\ell_{ij}),$$

$$\alpha' = (\alpha_2, \dots, \alpha_p) = -\phi_2' \Sigma_{22.1}^{-1/2}, \phi(t) = (2\pi)^{-1/2} \exp(-t^2/2),$$

$$\beta' = (\beta_2, \dots, \beta_p) = (R_2 - \phi_2)' \Sigma_{22.1}^{-1/2}, R' = (R_2, \dots, R_p), \phi'_2 = (\phi_2, \dots, \phi_p)$$

and ϕ_2 and $\Sigma_{22.1}$ denote the conditional mean and variance of vector y_2 given y_1 .

Multiplying (7) by R_1 and subtracting the resulting equation from (7) with $r_1 + 1$ in place of r_1 , we get the recurrence relation in moments as

$$\begin{aligned}
(9) \quad (r_1 + 1) \mu'_{r_1, \dots, r_p} - R_1 r_1 \mu'_{r_1-1, r_2, \dots, r_p} \\
&= \sum_{j=1}^p h_{ij} (\mu'_{r_1+1, \dots, r_j+1, \dots, r_p} - R_1 \mu'_{r_1, \dots, r_j+1, \dots, r_p}) \\
&+ \eta_{r_1+1, r_2, \dots, r_p} - R_1 \mu'_{r_1, \dots, r_p}.
\end{aligned}$$

A further $(p-1)$ recurrence relations similar to (9) with respect to $r_2, \dots, r_p$ can be written down. These p recurrence relations yield in all $p(p+1)$ equations in which the p -tuples $(r_1, \dots, r_p)$ satisfy the condition $\sum_{j=1}^p r_j \leq 2$.

3. Derivation of Estimators

We have just noted that there are $p(p+1)$ equations in all. We have a vector e of $s = p(p+3)/2$ parameters where $e' : 1xs = (h_{11}, h_{12}, \dots, h_{1p}, h_{22}, h_{23}, \dots, h_{pp}, \eta_1, \dots, \eta_p)$. We have to select s equations out of $p(p+1)$ equations. In the selection of the s equations, we include p equations corresponding to the p p -tuples $\{(1, 0, \dots, 0), \dots, (0, \dots, 0, 1)\}$ which give moments of smaller order and p equations corresponding to the p p -tuples $\{(2, 0, \dots, 0), \dots, (0, \dots, 0, 2)\}$ which provide information regarding the variance. Hence we have to select $s-2p = \frac{1}{2}p(p-1)$ equations from the remaining $p(p-1)$ equations. Thus the total number of sets of s equations each is $t = 2^{p(p-1)/2}$.

Choosing the i -th set out of t sets, we obtain the following system of linearly independent equations.

$$(10) \quad A_i e = C_i, i=1, \dots, t,$$

where $A_i : s \times s$ is a square matrix and C_i is an $s \times 1$ vector involving elements $\mu'_{r_1, \dots, r_p}$ with $(r_1, \dots, r_p)$ assuming different values corresponding to the i -th set.

If we replace, in (10), the population moments $\mu'_{r_1, \dots, r_p}$ by the corresponding

$$\text{sample moments } m'_{r_1, \dots, r_p} = n^{-1} \sum_{j=1}^n y_{1j}^{r_1} \dots y_{pj}^{r_p}$$

$= n^{-1} \sum_{j=1}^n (x_{1j} - a_{11})^{r_1} \dots (x_{pj} - a_{1p})^{r_p}$, the solution e (to be denoted by $\hat{e}_i$) is given by

$$(11) \quad \hat{e}_i = \hat{A}_i^{-1} \hat{C}_i, i=1, \dots, t$$

It may be noted that the solutions $\hat{e}_i$ for $i=1, \dots, t$ are not symmetric in $(x_1, \dots, x_p)$.

We now define the vector $\hat{\bar{e}}_1$ as an estimator of e by

$$(12) \quad \hat{\bar{e}}_1 = \sum_{i=1}^t \hat{e}_i / t$$

Since $\hat{\bar{e}}_1$ is symmetric in $(x_1, \dots, x_p)$, we recommend $\hat{\bar{e}}_1$ as an estimator of e . The estimates of μ and Σ obtained from $\hat{\bar{e}}_1$ may be used as a starting vector for solving the maxi-

mum likelihood equations and should, in general, reduce the number of cycles required to yield a solution. Similarly combining all the t equations in (10), we get a single equation

$$(13) \quad \bar{A} \hat{e} = \bar{C},$$

where

$$\bar{A} = \sum_{i=1}^t A_i/t \quad \text{and} \quad \bar{C} = \sum_{i=1}^t C_i/t.$$

Then we get the solution $\hat{e}_2$ of e as

$$(14) \quad \hat{e}_2 = \hat{A}^{-1} \hat{C}.$$

Note that $\hat{e}_2$ is also symmetric in observations. One can also recommend the vector $\hat{e}_3$ as an estimator for e defined by

$$(15) \quad \hat{e}_3 = \frac{1}{2}(\hat{e}_1 + \hat{e}_2).$$

Khatri and Jaiswal (1963) have shown that $\hat{e}_3$ is more efficient than $\hat{e}_1$, $\hat{e}_2$, $\hat{e}_1$ and $\hat{e}_2$ in the bivariate case.

In particular, for $p=2$ (i.e. $t=2$, the two sets being

$$\{(1,0), (2,0), (1,1), (0,1), (2,0)\} \quad \text{and} \quad \{(1,0), (2,0), (0,1), (2,0), (1,1)\}$$

the two solutions $\hat{e}_1$ and $\hat{e}_2$ given by (10) become

$$(16) \quad \hat{e}_1 = M^{-1} u,$$

and

$$(17) \quad \hat{e}_2 = N^{-1} v,$$

where

$$\hat{e}' = (h_{11}, h_{12},$$

$$\hat{e}' = (h_{11}, h_{12}, h_{22}, \hat{\eta}_1, \hat{\eta}_2),$$

$$M = \begin{pmatrix} m'_{30} - R_1 m'_{20} & m'_{21} - R_1 m'_{11} & 0 & m'_{20} - R_1 m'_{10} & 0 \\ m'_{40} - R_1 m'_{30} & m'_{31} - R_1 m'_{21} & 0 & m'_{30} - R_1 m'_{20} & 0 \\ m'_{31} - R_1 m'_{21} & m'_{22} - R_1 m'_{12} & 0 & m'_{21} - R_1 m'_{11} & 0 \\ 0 & m'_{12} - R_2 m'_{11} & m'_{03} - R_2 m'_{02} & 0 & m'_{02} - R_2 m'_{01} \\ 0 & m'_{13} - R_2 m'_{12} & m'_{04} - R_2 m'_{03} & 0 & m'_{03} - R_2 m'_{02} \end{pmatrix}$$

$$N = \begin{pmatrix} m'_{30} - R_1 m'_{20} & m'_{21} - R_1 m'_{11} & 0 & m'_{20} - R_1 m'_{10} & 0 \\ m'_{40} - R_1 m'_{30} & m'_{31} - R_1 m'_{21} & 0 & m'_{30} - R_1 m'_{20} & 0 \\ 0 & m'_{22} - R_2 m'_{21} & m'_{13} - R_2 m'_{12} & 0 & m'_{12} - R_2 m'_{11} \\ 0 & m'_{12} - R_2 m'_{11} & m'_{03} - R_2 m'_{02} & 0 & m'_{02} - R_2 m'_{01} \\ 0 & m'_{13} - R_2 m'_{12} & m'_{04} - R_2 m'_{03} & 0 & m'_{03} - R_2 m'_{02} \end{pmatrix}$$

$$u = \begin{pmatrix} 2m'_{10} - R_1 m'_{00} \\ 3m'_{20} - 2R_1 m'_{10} \\ 2m'_{11} - R_1 m'_{01} \\ 2m'_{01} - R_2 m'_{00} \\ 3m'_{02} - 2R_2 m'_{01} \end{pmatrix} \quad \text{and} \quad v = \begin{pmatrix} 2m'_{10} - R_1 m'_{00} \\ 3m'_{20} - R_1 m'_{10} \\ 2m'_{11} - R_2 m'_{10} \\ 2m'_{01} - R_2 m'_{00} \\ 3m'_{02} - 2R_2 m'_{01} \end{pmatrix}$$

It may be noted that the third equation obtained by Dyer (1973) in the system of linearly independent equations (given by (7) in Section 3 of his paper) can be obtained by taking the difference of (16) and (17) involving ξ_1 and ξ_2 . However there is no unique way of obtaining the similar equation when we consider the p-variate normal distribution.

4. Asymptotic Variances of the Estimators

Denoting by R^* the differential of R and noting that $(R_1 R_2)^* = R_1^* R_2 + R_1 R_2^*$, from $A_i e_i = C_i$ for $i=1, 2, \dots, t$, we get

$$(18) \quad e_i^* = A_i^{-1} (C_i^* - A_i^* e_i),$$

where A_i, e_i are matrices in the population parameters and differentials are taken with respect to moments. Hence, using the differential method given by Kendall and Stuart (1963, pp. 231-236) and the recurrence relations given by (9), we can write the covariance matrix of the estimators of e_i as

$$(19) \quad V(\xi_i) = A_i^{-1} V A_i^{-1'}, i=1, 2, \dots, t$$

where

$$V = E \{ (C_i^* - A_i^* e_i)(C_i^* - A_i^* e_i)' \}$$

However, further expressions for V are quite cumbersome and will not be given here.

Since sample moments converge in probability to the corresponding population moments, by Slutsky's theorem (cf. Cramér (1946), p. 255) any rational function of

the sample moments converges in probability to the same rational function of population moments. It follows that the rational functions of the estimators given by (10) are consistent.

5. Method of Maximum Likelihood

The logarithm of the likelihood based on a random sample of size n from (1) is given by

$$(20) \quad \text{Log}L = \text{Constant} - n \log G - \frac{n}{2} \log |\Sigma| - \frac{1}{2} \sum_{\alpha=1}^n (x_{\alpha} - \mu)' \Sigma^{-1} (x_{\alpha} - \mu)$$

where G is the same as defined in (3) of Section 2.

Differentiating (20) with respect to μ_i , σ_{ii} , σ_{ij} ($i \neq j$) and equating to zero, we get the likelihood equations as

$$(21) \quad \frac{\partial \log L}{\partial \mu_i} = \frac{-n}{G} \frac{\partial G}{\partial \mu_i} + \sum_{\alpha=1}^n \sigma_{ii} (x_{\alpha i} - \mu_i) + \sum_{\alpha=1}^n \sum_{j=1, j \neq i}^p \sigma_{ij} (x_{\alpha j} - \mu_j),$$

$$(22) \quad \frac{\partial \log L}{\partial \sigma_{ii}} = \frac{-n}{G} \frac{\partial G}{\partial \sigma_{ii}} - \frac{n}{2} \sigma_{ii} + \frac{n}{2} \text{tr} \left[\Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_{ii}} \Sigma^{-1} \{S + (\bar{x} - \mu)(\bar{x} - \mu)'\} \right]$$

$$(23) \quad \frac{\partial \log L}{\partial \sigma_{ij}} = \frac{-n}{G} \frac{\partial G}{\partial \sigma_{ij}} - n \sigma_{ij} + \frac{n}{2} \text{tr} \left[\Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_{ij}} \Sigma^{-1} \{S + (\bar{x} - \mu)(\bar{x} - \mu)'\} \right]$$

where

$$(24) \quad \bar{x}_j = \sum_{\alpha=1}^n x_{\alpha j} / n, \quad \bar{x}' = (\bar{x}_1, \dots, \bar{x}_p),$$

$$S = \frac{1}{n} \sum_{\alpha=1}^n (x_{\alpha} - \bar{x})(x_{\alpha} - \bar{x})'$$

Differentiating G , w.r.t. μ_i , we have

$$\begin{aligned} \frac{\partial G}{\partial \mu_i} &= - \int_{a_1}^{a_2} \frac{\partial g}{\partial x_i} (x|\mu, \Sigma) dx \\ &= \frac{1}{\sqrt{\sigma_{ii}}} [\phi(\xi_{1i}) \int_{\xi_{1(i)}}^{\xi_{2(i)}} g(u_{(i)}/u_i = \xi_{1i}) du_{(i)} \\ &\quad - \phi(\xi_{2i}) \int_{\xi_{1(i)}}^{\xi_{2(i)}} g(u_{(i)}/u_i = \xi_{2i}) du_{(i)}] \end{aligned}$$

where

$$u_j = (x_j - \mu_j)/\sqrt{\sigma_{jj}}, \xi_{ij} = (a_{ij} - \mu_j)/\sqrt{\sigma_{jj}}, j = 1, 2, \dots, p, i = 1, 2$$

$$u'_{(i)} = (u_1, \dots, u_{i-1}, u_{i+1}, \dots, u_p) \text{ and}$$

$$\xi'_{r(i)} = (\xi_{r1}, \dots, \xi_{ri-1}, \xi_{ri+1}, \dots, \xi_{rp}) \text{ for } r = 1, 2.$$

We can write $\frac{\partial G}{\partial \mu_i}$ as

$$(25) \quad \frac{\partial G}{\partial \mu_i} = \frac{G P_i}{\sqrt{\sigma_{ii}}},$$

where

$$(26) \quad \left\{ \begin{array}{l} P_i = P_{11.i} - P_{12.i} - P_{21.i} + P_{22.i} \\ P_{rS.i} = \phi(\xi_{ri}) I(\xi_{S(i)} | \xi_{ri}) / G \\ \text{and} \\ I(\xi_{S(i)} | \xi_{ri}) = \int_{\xi_{S(i)}}^{\infty} g(u_{(i)} | u_i = \xi_{ri}) du_{(i)} \text{ for } r, S = 1, 2. \end{array} \right.$$

Again differentiating G , w.r.t. σ_{ii} , we get

$$\begin{aligned} \frac{\partial G}{\partial \sigma_{ii}} &= \frac{1}{2} \int_{a_1}^{a_2} \frac{\partial^2 g(x | \mu, \Sigma)}{(\partial x_i)^2} dx \\ &= -\frac{1}{2} \int_{a_1}^{a_2} \frac{\partial}{\partial x_i} [(x_i - \mu_i) \sigma^{ii} g(x | \mu, \Sigma)] dx \\ &= -\frac{\sigma^{ii} G}{2} [\xi_{2i} P_{21.i} - \xi_{2i} P_{22.i} - \xi_{1i} P_{11.i} + \xi_{1i} P_{12.i}] \end{aligned}$$

where $P_{rS.i}$ for $r, S = 1, 2$ are the same as in (26). We can write

$$(27) \quad \frac{\partial G}{\partial \sigma_{ii}} = \frac{\sigma^{ii} G \xi_i P_i}{2}$$

where

$$(28) \quad \xi_i P_i = \xi_{1i} P_{11.i} - \xi_{1i} P_{12.i} - \xi_{2i} P_{21.i} + \xi_{2i} P_{22.i}$$

Finally, according to Basu and Khatri (1969):

$$\frac{\partial G}{\partial \sigma_{ij}} = \int_{a_1}^{a_2} \frac{\partial^2 g(x | \mu, \Sigma)}{\partial x_i \partial x_j} dx, \text{ which yields}$$

$$(29) \quad \frac{\partial G}{\partial \sigma_{ij}} = \int_{a_{1(ij)}}^{a_{2(ij)}} g(x_i)g(x_j|x_i)g(x_{(ij)}|x_i, x_j) \left. \begin{matrix} x_i=a_{2i} \\ x_j=a_{2j} \\ x_i=a_{1i} \\ x_j=a_{1j} \end{matrix} \right\} dx_{(ij)}$$

where $a_{1(ij)}$ and $a_{2(ij)}$ are obtained from a_1 and a_2 by deleting (a_{1i}, a_{1j}) and (a_{2i}, a_{2j}) .

Similarly $x_{(ij)}$ is defined from x .

We can write (29) as

$$\begin{aligned} \frac{\partial G}{\partial \sigma_{ij}} &= \frac{\phi(\xi_{2j})}{\sqrt{\sigma_{ii}\sigma_{jj}(1-\rho_{ij}^2)}} [\phi(\xi_{2j}|\xi_{2i}) \int_{a_{1(ij)}}^{a_{2(ij)}} g(x_{(ij)} | x_i = a_{2i}, x_j = a_{2j}) dx_{(ij)} \\ &\quad - \phi(\xi_{1j}|\xi_{2i}) \int_{a_{1(ij)}}^{a_{2(ij)}} g(x_{(ij)} | x_i = a_{2i}, x_j = a_{1j}) dx_{(ij)}] \\ &\sim \frac{\phi(\xi_{1i})}{\sqrt{\sigma_{ii}\sigma_{jj}(1-\rho_{ij}^2)}} [\phi(\xi_{2j}|\xi_{1i}) \int_{a_{1(ij)}}^{a_{2(ij)}} g(x_{(ij)} | x_i = a_{1i}, x_j = a_{2j}) dx_{(ij)} \\ &\quad - \phi(\xi_{ij}|\xi_{1i}) \int_{a_{1(ij)}}^{a_{2(ij)}} g(x_{(ij)} | x_i = a_{1i}, x_j = a_{1j}) dx_{(ij)}] \end{aligned}$$

We write $\partial G/\partial \sigma_{ij}$ as

$$(30) \quad \frac{\partial G}{\partial \sigma_{ij}} = \frac{\rho_{ij} G A_{ij}}{\sqrt{\sigma_{ii}\sigma_{jj}(1-\rho_{ij}^2)}},$$

where

$$(31) \quad \left\{ \begin{aligned} A_{ij} &= [A_{22,ij} I_{22,ij} - A_{12,ij} I_{12,ij} - A_{21,ij} I_{21,ij} + A_{11,ij} I_{11,ij}] \\ A_{rS,ij} &= \phi(\xi_{ri}) \phi(\xi_{sj}|\xi_{ri}) \\ \text{and} \\ I_{rS,ij} &= \int_{a_{1(ij)}}^{a_{2(ij)}} g(x_{(ij)} | x_i = a_{ri}, x_j = a_{sj}) dx_{(ij)} / G \end{aligned} \right.$$

for $r, S = 1, 2$.

Using (25), (27) and (30), the likelihood equations (21)-(23) can be written as

$$(32) \quad \frac{\partial \log L}{\partial \mu_i} = \frac{-n P_i}{\sqrt{\sigma_{ii}}} + n \sigma_{ii} (\bar{x}_i - \mu_i) + n \sum_{j=1}^p \sigma^{ij} (\bar{x}_j - \mu_j),$$

$$(33) \quad \frac{\partial \log L}{\partial \sigma_{ii}} = \frac{-n \sigma_{ii} [1 + \xi_i P_i]}{2} + \frac{n}{2} \text{tr} \left[\Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_{ii}} \Sigma^{-1} \{S + (\bar{x} - \mu)(\bar{x} - \mu)'\} \right]$$

and

$$(34) \quad \frac{\partial \log L}{\partial \sigma_{ij}} = -n \sigma^{ij} - \frac{n \rho_{ij} A_{ij}}{\sqrt{\sigma_{ii} \sigma_{jj} (1 - \rho_{ij}^2)}} + \frac{n}{2} \text{tr} \left[\Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_{ij}} \Sigma^{-1} \{S + (\bar{x} - \mu)(\bar{x} - \mu)'\} \right]$$

for $i = 1, \dots, p$ and $i \neq j = 1, \dots, p$.

In particular we show that the likelihood equations obtained by Nath (1971) can be deduced from equations (32), (33) and (34).

Taking $p = 2$ and noting that $\sigma^{ii} = 1 / \{\sigma_{ii}(1 - \rho^2)\}$ for $i = 1, 2$ and $\sigma^{12} = -\rho / \{\sqrt{(\sigma_{11}\sigma_{22})(1 - \rho^2)}\}$, equations (32), reduce to

$$-\hat{P}_1 + \frac{(\bar{x}_1 - \hat{\mu}_1)}{\hat{\sigma}_1(1 - \hat{\rho}^2)} - \frac{\hat{P}(\bar{x}_2 - \hat{\mu}_2)}{\hat{\sigma}_2(1 - \hat{\rho}^2)} = 0$$

and

$$-\hat{P}_2 - \frac{\hat{\rho}(\bar{x}_1 - \hat{\mu}_1)}{\hat{\sigma}_1(1 - \hat{\rho}^2)} + \frac{(\bar{x}_2 - \hat{\mu}_2)}{\hat{\sigma}_2(1 - \hat{\rho}^2)} = 0,$$

where $\sigma_{ii} = \sigma_i^2$, $i = 1, 2$.

Equivalently we obtain the estimates of μ_1 and μ_2 :

$$(35) \quad \hat{\mu}_1 = \bar{x}_1 - (\hat{P}_1 + \hat{\rho} \hat{P}_2) \hat{\sigma}_1$$

$$\hat{\mu}_2 = \bar{x}_2 - (\hat{P}_2 + \hat{\rho} \hat{P}_1) \hat{\sigma}_2,$$

which are the same as those given by Nath (1971).

Similarly, using the relations

$$\frac{\partial G}{\partial \sigma_{11}} = \frac{G}{2\sigma_1^2(1 - \rho^2)} \xi_1 P_1, \quad \frac{\partial G}{\partial \sigma_{22}} = \frac{G}{2\sigma_2^2(1 - \rho^2)} \xi_2 P_2 \quad \text{and}$$

$$\frac{\partial G}{\partial \sigma_{12}} = \frac{G\rho}{\sigma_1\sigma_2(1-\rho^2)} A_{12}$$

in equations (33) and (34) and after simplification we get the equations

$$\begin{aligned} \hat{\sigma}_1 &= S_1 [\hat{\rho}^2 (\hat{A}_{12} + \hat{\xi}_2 \hat{P}_2) + (1 + \hat{\xi}_1 \hat{P}_1) - (\hat{P}_1 + \hat{\rho} \hat{P}_2)^2]^{-1/2}, \\ (36) \quad \hat{\sigma}_2 &= S_2 [\hat{\rho}^2 (\hat{A}_{12} + \hat{\xi}_1 \hat{P}_1) + (1 + \hat{\xi}_2 \hat{P}_2) - (\hat{P}_2 + \hat{\rho} \hat{P}_1)^2]^{-1/2}, \\ \hat{\rho} &= rS_1 S_2 (\hat{\sigma}_1 \hat{\sigma}_2)^{-1} \{ (\hat{A}_{12} + \hat{\xi}_2 \hat{P}_2 + \hat{\xi}_1 \hat{P}_1 + 1) - \frac{1}{\hat{\rho}} (\hat{P}_1 + \hat{\rho} \hat{P}_2)(\hat{P}_2 + \hat{\rho} \hat{P}_1) \}^{-1}. \end{aligned}$$

which are again the same as those given by Nath (1971).

6. Numerical Example

In this section, we compare the estimates of the parameters of the doubly truncated bivariate normal distribution obtained from $\hat{\epsilon}_1, \hat{\epsilon}_2, \hat{\epsilon}_1, \epsilon_D$ (due to Dyer) and the maximum likelihood estimates using the method of iteration suggested by Nath (1971). The data are taken from the statistical summary of Medical Examination Reports (cf. Anonymous (1960), Table XXII, p. 35). The height below 60 inches and above 68 inches and mass below 85 lbs and above 130 lbs are truncated.

Thus we have

$$a_1 = \begin{pmatrix} 60 \\ 85 \end{pmatrix} \quad a_2 = \begin{pmatrix} 68 \\ 130 \end{pmatrix}.$$

Sample moments about the lower truncation points were found to be

$m'_{00} = 1,000,000$	$m'_{10} = 4,444.44$	$m'_{20} = 23,863.25$
$m'_{30} = 140,803.4$	$m'_{40} = 884,478.6$	$m'_{01} = 20,683.76$
$m'_{11} = 102,393.2$	$m'_{21} = 589,145.3$	$m'_{31} = 3658,547$
$m'_{02} = 552,991.5$	$m'_{12} = 2891,453$	$m'_{22} = 17253.42$
$m'_{03} = 16863.25$	$m'_{13} = 91277.78$	$m'_{04} = 560106.8$

Substituting these values in M and u and in N and v, defined in (16) and (17) we get two sets of equations. Solving the equations in the first set we get the solution vectors $\hat{\epsilon}' = (\hat{h}_{11}, \hat{h}_{12}, \hat{h}_{22}, \hat{\eta}_1, \hat{\eta}_2)$ as

$$\begin{aligned} \hat{\epsilon}'_1 &= (-0.61821; \quad 0.084768; \quad -0.021722; \quad 0.59548; \quad 0.10560) \\ \text{and} \quad \hat{\epsilon}'_2 &= (-0.59818; \quad 0.081258; \quad -0.021023; \quad 0.59035; \quad 0.10495). \end{aligned}$$

Furthermore using (6), we first compute $\hat{\sigma}_1$, $\hat{\sigma}_2$ and $\hat{\rho}$ and then $\hat{\mu}_1$ and $\hat{\mu}_2$.

Similarly we obtain the second set of estimates using N and ν . The third set of estimates are obtained by considering the average of $\hat{e}_1$ and $\hat{e}_2$.

Dyer's estimates are obtained by changing the third equation of the sets. Dyer's third equation can be obtained by taking the difference of the third equation of the two sets given by (16) and (17). The results so obtained are given in Table 1.

Computation of Maximum Likelihood Estimators is found to be laborious, as it requires computation of the probability of a bivariate normal distribution and further involves several iterations. We started with the sample values ($\bar{x}_1$, $\bar{x}_2$, S_1 , S_2 , r) as initial estimates of $(\mu_1, \mu_2, \sigma_1, \sigma_2, \rho)$. Using these estimates

$$G = \int_{a_1}^{a_2} g(x) dx_1 dx_2$$

was computed with the help of Owen's tables (1962) and G was found to be 0.9172135.

Solving equations (35) and (36) we get the first set of estimates as (63,7958; 102,0306; 2,390; 13,4030; 0,8675). After four cycles of iterations the solution converges to the estimates which are given in Table 1.

TABLE 1
DIFFERENT ESTIMATES OF THE PARAMETERS OF THE TRUNCATED BIVARIATE DISTRIBUTION FOR NUMERICAL EXAMPLES

Method

$\hat{e}_1$	63,5057	103,5421	1,8653	9,9510	0,7315
$\hat{e}_2$	63,5216	103,5681	1,8761	10,0074	0,7246
$\hat{e}_1$	63,5114	103,5531	1,8706	9,9791	0,7281
$\hat{e}_D$	62,1846	104,3132	2,0505	10,9151	0,6142
MLE	64,4403	104,7963	2,5795	14,2362	0,5847

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REFERENCES

ANONYMOUS, (1960) *Statistical Summary of Medical Examination Reports of the Students of M S University of Baroda for 1958-1959*. M S University of Baroda, Faculty of Science. Baroda.

- BASU, D. and KHATRI, C.G. (1960). On some characterizations of statistics. *Sankhya* Series A, 32 (Part II), p. 203-204.
- CRAMÉR, H. (1946). *Mathematical methods of statistics*, Princeton University Press, Princeton.
- DYER, D.D. (1973). On moments estimation of the parameters of a truncated bivariate normal distribution, *Applied Statistics* 22 287-291.
- KENDALL, M.G. and STUART, A. (1963). *The advanced theory of Statistics*, Vol. I (Second Edition), Charles Griffin and Company, London.
- KHATRI, C.G. and JAISWAL, M.C. (1963). Estimation of parameters of a truncated bivariate normal distribution. *Jour. Amer. Statist. Associ.* 58 519-526.
- NATH, G.B. (1971). Estimation of truncated bivariate normal distribution. *Applied Statistics* 20 313-319.
- OWEN, D.B. (1962). *Handbook of Statistical Tables*, Addison Wesley Publishing Co., Inc., Reading, Massachusetts.

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